JS Arrays

Shoaib Ghachi

MCT | Technical Trainer | Mentor |SME

Overview

JavaScript Arrays

- -Creating an Array
- -Accessing Elements
- -Modifying Elements
- -Array Length
- -Adding Elements
- -Removing Elements
- -Iterating Over an Array

JS Arrays

In JavaScript, numbers are used to represent numeric values. They can be integers, floating-point numbers, or special numeric values like NaN (Not-a-Number) and Infinity. Here's an overview of working with numbers in JavaScript:

- **Creating an Array:**

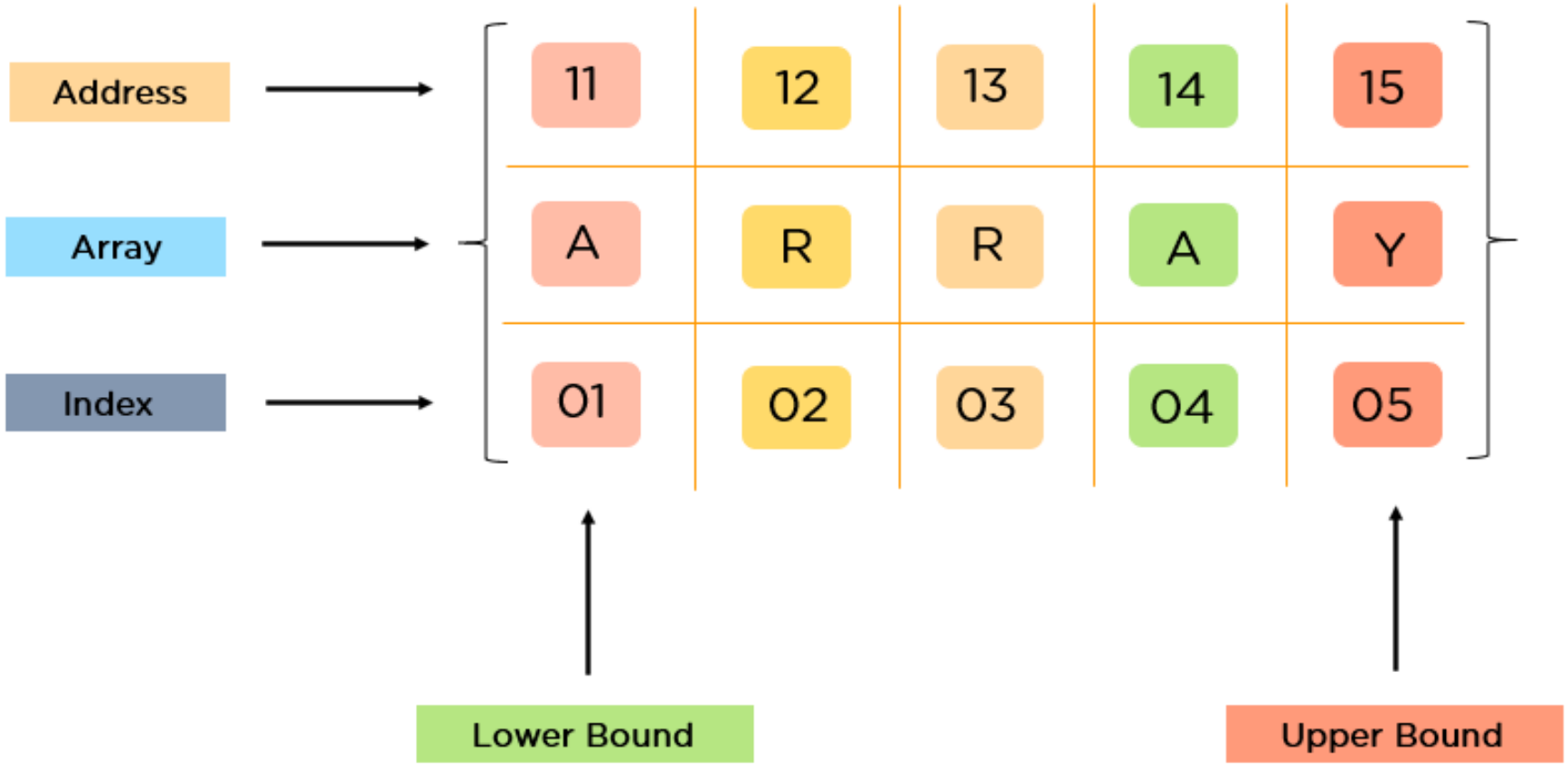
You can create an array using array literals (square brackets) or the Array constructor:

```
const array1 = [1, 2, 3, 4, 5]; // Using array literals
const array2 = new Array('apple', 'banana', 'orange'); // Using Array constructor
```

- **Accessing Elements:**

You can access individual elements of an array using their index, starting from 0:

```
const array = ['apple', 'banana', 'orange'];
console.log(array[0]); // Output: 'apple'
console.log(array[1]); // Output: 'banana'
```


Address
Array
Index
11
12
13
14
15
A
R
R
A
Y
01
02
03
04
05
Lower Bound
Upper Bound

- **Modifying Elements:**

You can modify the value of an element by assigning a new value to a specific index:

```
const array = ['apple', 'banana', 'orange'];
array[1] = 'grape';
console.log(array); // Output: ['apple', 'grape', 'orange']
```

- **Array Length:**

The length property gives you the number of elements in an array:

```
const array = [1, 2, 3, 4, 5];
console.log(array.length); // Output: 5
```

- **Adding Elements:**

You can add elements to the end of an array using the push() method:

```
const array = ['apple', 'banana', 'orange'];
array.push('grape');
console.log(array); // Output: ['apple', 'banana', 'orange', 'grape']
```

- **Removing Elements:**

You can remove elements from an array using the pop() method to remove the last element or the splice() method to remove elements at specific positions:

```
const array = ['apple', 'banana', 'orange'];
array.pop(); // Removes the last element
console.log(array); // Output: ['apple', 'banana']

const array2 = ['apple', 'banana', 'orange'];
array2.splice(1, 1); // Removes one element at index 1
console.log(array2); // Output: ['apple', 'orange']
```

- **Iterating Over an Array:**

You can use loops or array methods like forEach() or map() to iterate over the elements of an array:

```
const array = ['apple', 'banana', 'orange'];
// Using for loop
for (let i = 0; i < array.length; i++) {
  console.log(array[i]);
}
// Using forEach method
array.forEach(function(element) {
  console.log(element);
});
// Using map method
const newArray = array.map(function(element) {
  return element.toUpperCase();
});
console.log(newArray); // Output: ['APPLE', 'BANANA', 'ORANGE']
```

References

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